Measuring Delivery Delays in Online Retail

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Abstract

We study how long parcels take to arrive and how often they are lost. Late deliveries are defined as arriving more than three days after the estimate.

1 Introduction

Online retailers promise delivery windows that customers rely on. When a parcel is late, the customer contacts support, and the cost of that contact often exceeds the margin on the order. This paper measures delays across carriers and seasons. We find that most delays cluster in December, when volumes double and sorting centres run at capacity.

2 Data

The dataset covers 120,000 shipments sent between January and December. Each record contains the dispatch date, the promised window, the delivery date and the carrier. Shipments without a delivery scan were treated as lost after fifteen working days.

2.1 Cleaning

We removed duplicate scans and shipments with impossible dates. About two per cent of records were excluded in total.

3 Results

The median delay was zero days, but the distribution has a long tail: five per cent of parcels arrived more than a week late. Economy services were four times as likely to be late as tracked services. Losses were rare, at roughly one parcel in two hundred.

4 Topic 5

This section discusses topic number 5 in some depth. Sentence 1 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 2 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 3 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 4 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 5 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 6 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 7 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year. Sentence 8 of topic 5 explains the measurements, the carriers involved and the seasonal effects observed across the year.

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11 Conclusion

Retailers can reduce support contacts by sending proactive delay notices in December and by steering high-value orders towards tracked services.